DB-Curator

Agentic database construction for scientific evidence

A tool that searches literature, proposes candidates, and verifies each database row with auditable evidence

Chen Zhao, Botao Liu, Xiaoxiao Li, Tao Huan

Background: scientific databases are reusable knowledge infrastructure

Scientific databases turn scattered papers into searchable, comparable and reusable knowledge.

1. Define scope

Experts decide what should count as an entry.

Borderline cases need explicit inclusion and exclusion rules.

2. Search papers

Relevant evidence is scattered across many articles.

The literature set changes as the topic becomes clearer.

3. Interpret context

A name mention is not enough.

The row needs identity, matrix, mechanism and evidence context.

4. Record evidence

Accepted entries need citations and provenance.

Future users must be able to inspect why the row exists.

Takeaway: scientific curation is slow because every accepted row needs identity, scope and evidence provenance.

Problem: extraction and RAG do not decide database membership

A chemical mention can be a target compound, reagent, precursor, solvent, contaminant or contextual comparison.

Standard extraction / RAG

Finds entity mentions or retrieves passages.

Generates an answer from a context window.

Citations can look plausible even when they do not support the exact claim.

The output is usually a one-shot answer.

Scientific database construction

Decides whether an entity deserves a row.

Assigns an entry-level verification state.

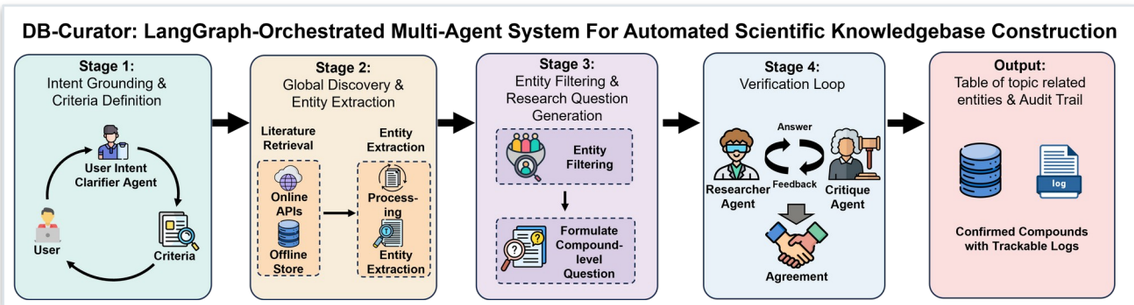
The cited paper must support the exact rubric predicate.

The output is an inspectable, correctable database artifact.

missing evidence | exclusion evidence | off-scope

These should change the row label, not just change answer wording.

Core idea: turn user intent into an evidence contract



1. Dynamic literature construction

Search changes as candidates and evidence gaps appear.

2. Candidate grounding

Names are normalized and filtered before expensive verification.

3. Predicate-state verification

Researcher finds evidence; Critique checks support and exclusions.

supported **unresolved** **exclusion** **exhausted**

Accepted = required predicates supported + no triggered exclusion.

Application demo: DBP database construction

User request: build a database of experimentally confirmed disinfection byproducts.

Refined rubric

Water-relevant disinfection or oxidation context.

Formation from disinfectant or oxidant reaction with relevant precursors.

Structural confirmation in peer-reviewed studies.

Agent execution

Expand DBP-specific literature searches.

Extract and ground chemical entities.

Ask compound-level verification questions.

Example question

Is NDMA an experimentally confirmed DBP?

Which paper supports formation, matrix and analytical confirmation?

Which contexts should block release?

label

Confirmed / limited / negative

evidence

quotes + citations

state

predicate audit trail

logs

search history

Verification: how do we know the output is reliable?

Paper-level audit

Does the cited item resolve to a real primary paper?

Does the paper contain the quoted span or equivalent local context?

Compound lookup pages support identity grounding, but are not counted as primary evidence.

Predicate-level audit

The paper must support the actual rubric predicate.

Mentioning the same compound is not enough.

Reagent, solvent, precursor or out-of-scope context can block release.

2,166/2,166

fixed-cohort support

Refs with quote/context support.

7,430/7,453

benchmark-scale support

Auditable refs across five domains.

11/257

zero-shot baseline

Usable refs from direct generation.

Trust argument: the output is not trusted because the model says so; it is trusted because the cited paper and predicate can be inspected.

Results and ML lesson: why agentic design matters

Release-ready construction

246/250 release-ready rows

68 benchmark-missing, human-validated entries

4 rejected non-overlap errors

Benchmark-scale verification

1,887 evaluated candidates across five domains

977 system Confirmed labels

7,430/7,453 auditable primary-paper refs

Critique-loop utility

49 evidence-limited rows rescued to Confirmed

5 premature Confirmed labels tightened away

51/54 changed decisions judged right by human inspection

Final ML lesson: the value of agentic AI in science is not fluent generation. It is dynamic evidence construction, explicit state tracking and critique-driven correction.

Practical implication: humans audit compact evidence trails and edge cases instead of manually searching every paper from scratch.